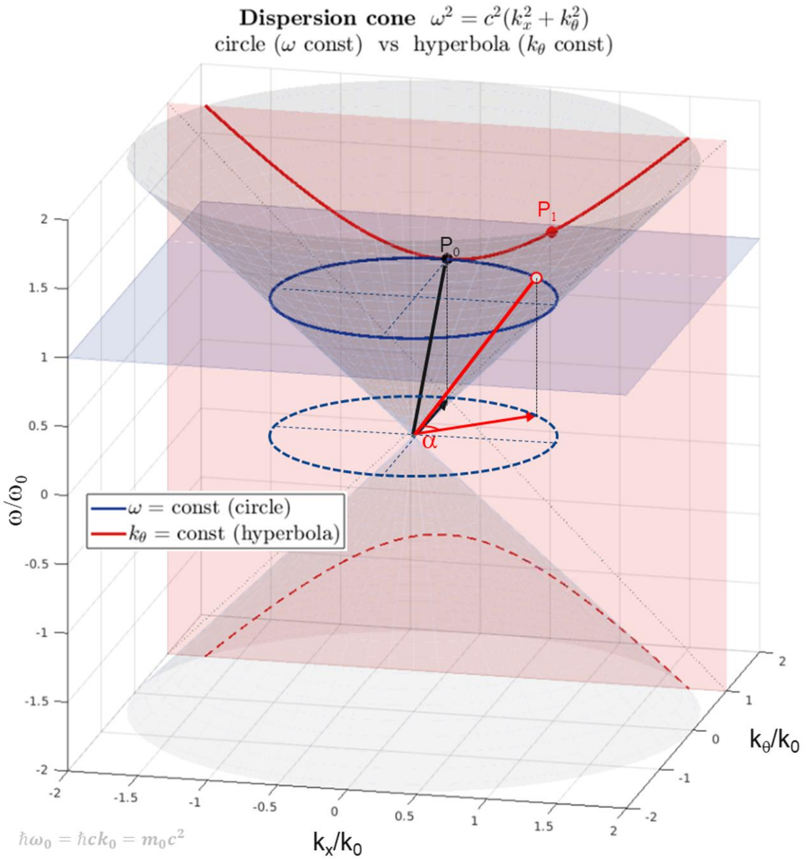

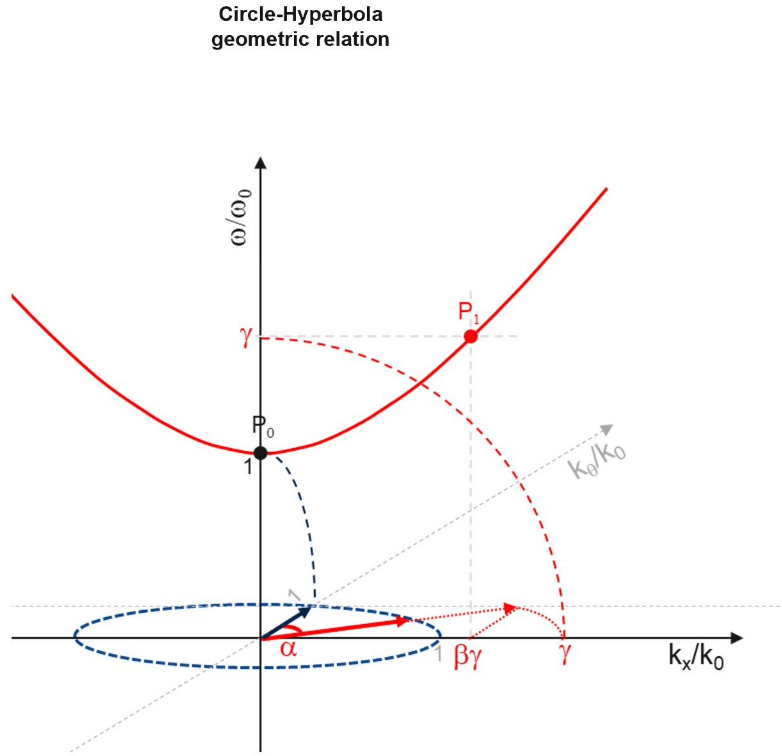

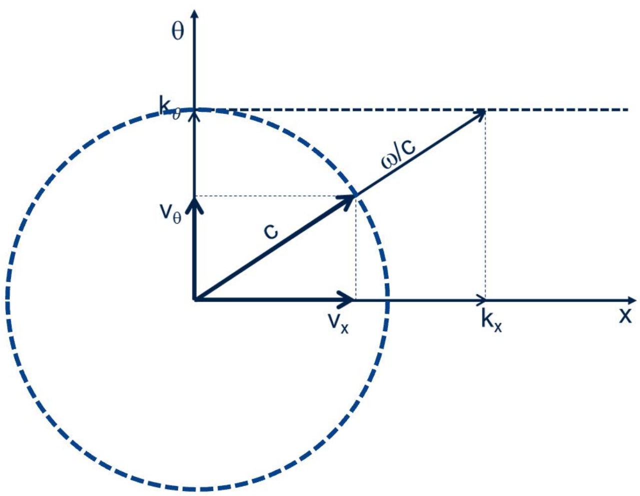

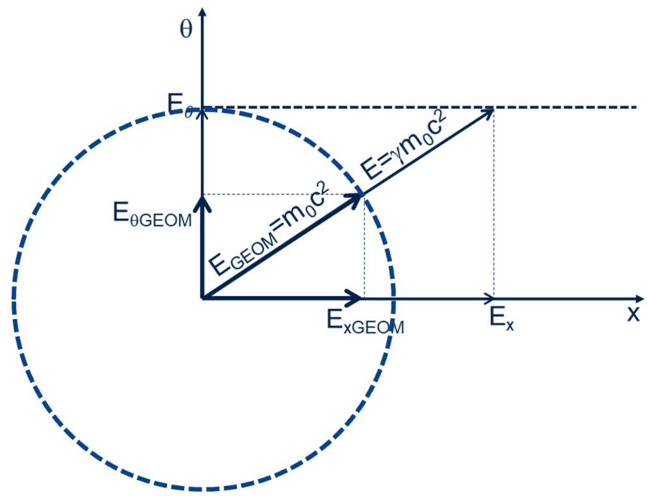

$\boldsymbol{\omega^2 = c^2\left(k_\theta^2 + k_x^2\right)}$

$\boldsymbol{k_\theta = \dfrac{2\pi}{\lambda_0} = \dfrac{m_0 c}{\hbar}}$ ***invariant***

$\boldsymbol{E^2 = E_\theta^2 + E_x^2}$

$\boldsymbol{E_\theta = m_0 c^2}$ ***invariant***

Hyperbola

$$v_\theta^2 = \left(\frac{\delta\omega}{\delta k_\theta}\right)^2 = \frac{k_\theta^2}{k_\theta^2 + k_x^2} c^2 = \frac{k_\theta^2}{\omega^2} c^4$$

$$v_x^2 = \left(\frac{\delta\omega}{\delta k_x}\right)^2 = \frac{k_x^2}{k_\theta^2 + k_x^2} c^2 = \frac{k_x^2}{\omega^2} c^4$$

$E_x = p_x c$

$E = \gamma m_0 c^2$

$E_{xGEOM} = \beta m_0 c^2$

$E_{GEOM} = m_0 c^2$

$\boldsymbol{c^2 = v_\theta^2 + v_x^2}$

c invariant

$\boldsymbol{E_{GEOM}^2 = E_{\theta GEOM}^2 + E_{xGEOM}^2}$

$\boldsymbol{E_{GEOM} = m_0 c^2}$ ***invariant***

Circle